

For GNSS setup:

```
File Edit Tabs Help
pi@raspberrypi:~$ source gnss_env/bin/activate
(gnss_env) pi@raspberrypi:~$ sudo apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Hit:4 http://archive.raspberrypi.com/debian trixie InRelease
218 packages can be upgraded. Run 'apt list --upgradable' to see them.
(gnss_env) pi@raspberrypi:~$ sudo apt install usb-modeswitch usb-modeswitch-data
usb-modeswitch is already the newest version (2.6.1-4+b2).
usb-modeswitch-data is already the newest version (20191128-7).
usb-modeswitch-data set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 218
(gnss_env) pi@raspberrypi:~$ sudo modprobe option
(gnss_env) pi@raspberrypi:~$ sudo modprobe usbserial
(gnss_env) pi@raspberrypi:~$ echo "2c7c 0904" | sudo tee /sys/bus/usb-serial/drivers/option1/new_id
2c7c 0904
(gnss_env) pi@raspberrypi:~$ dmesg | tail -20
[ 993.663068] usb 1-1.3: config 1 has an invalid interface number: 8 but max is 7
[ 993.663090] usb 1-1.3: config 1 has no interface number 6
[ 993.663198] usb 1-1.3: config 1 interface 0 altsetting 0 endpoint 0x81 has an invalid bInterval 32, changing to 9
[ 993.663502] usb 1-1.3: New USB device found, idVendor=2c7c, idProduct=0904, bcdDevice= 3.18
[ 993.663522] usb 1-1.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3
[ 993.663535] usb 1-1.3: Product: Android
[ 993.663546] usb 1-1.3: Manufacturer: Android
[ 993.675039] cdc_ether 1-1.3:1.0 usb0: register 'cdc_ether' at usb-0000:01:00.0-1.3, CDC Ethernet Device, 02:4b:b3:b9:eb:e5
[ 993.677528] option 1-1.3:1.2: GSM modem (1-port) converter detected
[ 993.678388] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB0
[ 993.679795] option 1-1.3:1.3: GSM modem (1-port) converter detected
[ 993.680808] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB1
[ 993.682221] option 1-1.3:1.4: GSM modem (1-port) converter detected
[ 993.682624] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB2
[ 993.683635] option 1-1.3:1.5: GSM modem (1-port) converter detected
[ 993.684128] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB3
[ 993.684916] option 1-1.3:1.7: GSM modem (1-port) converter detected
[ 993.685310] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB4
[ 993.686149] option 1-1.3:1.8: GSM modem (1-port) converter detected
[ 993.686511] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB5
(gnss_env) pi@raspberrypi:~$ ls /dev/ttyUSB*
/dev/ttyUSB0 /dev/ttyUSB2 /dev/ttyUSB4
/dev/ttyUSB1 /dev/ttyUSB3 /dev/ttyUSB5
(gnss_env) pi@raspberrypi:~$
```

AT commands:

Sent through Minicom:

```
Welcome to minicom 2.10

OPTIONS: I18n
Port /dev/ttyUSB0, 17:35:23 [U]

Press CTRL-A Z for help on special keys

OK

OK

+QGPSGNMEA: $GPGSV,3,1,11,41,75,167,44,15,69,088,45,29,68,353,41,45,67,135,43,1*6E
+QGPSGNMEA: $GPGSV,3,2,11,40,57,245,34,20
OK

+QGPSLOC: 120846.000,13.01070,80.23495,1.2,6.0,3,000.00,0.0,0.0,170326,15
```

AT commands:

AT

Turn ON GNSS: AT+QGPS=1

Check satellites: AT+QPSGNMEA="GSV"

Get location: AT+QGPSLOC=2

For LTE Setup:

```
pi@raspberrypi:~$ sudo systemctl enable ModemManager
pi@raspberrypi:~$ sudo systemctl start ModemManager
pi@raspberrypi:~$ sudo systemctl enable NetworkManager
pi@raspberrypi:~$ sudo systemctl start NetworkManager
pi@raspberrypi:~$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 2109:3431 VIA Labs, Inc. Hub
Bus 001 Device 010: ID 2c7c:0904 Quectel Wireless Solutions Co., Ltd. Android
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
pi@raspberrypi:~$ sudo modprobe option
pi@raspberrypi:~$ sudo modprobe usb_wwan
pi@raspberrypi:~$ echo "2c7c 0904" | sudo tee /sys/bus/usb-serial/drivers/option1/new-id
tee: /sys/bus/usb-serial/drivers/option1/new-id: Permission denied
2c7c 0904
pi@raspberrypi:~$ ls /dev/ttyUSB*
/dev/ttyUSB0 /dev/ttyUSB2 /dev/ttyUSB4
/dev/ttyUSB1 /dev/ttyUSB3 /dev/ttyUSB5
pi@raspberrypi:~$ sudo mmcli -L
/org/freedesktop/ModemManager1/Modem/0 [Quectel] EC200G
pi@raspberrypi:~$ nmcli con show
NAME                UUID                                  TYPE      DEVICE
netplan-wlan0-p1    acfb25b1-cbcd-3c83-8ca6-15f2af0ab563 wlan0
lte                  2dd7f195-c037-4274-93a1-20695d17d8eb gsm        ttyUSB5
lo                   e48d9338-1464-4748-a114-8e4782c1d0e5 loopback   lo
netplan-eth0        75a1216a-9d1a-30cd-8aca-ace5526ec021 ethernet  --
pi@raspberrypi:~$ sudo nmcli con up lte
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
pi@raspberrypi:~$ ping -c 4 google.com
PING google.com (2404:6800:4007:804::200e) 56 data bytes
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=1 ttl=116 time=27.8 ms
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=2 ttl=116 time=25.3 ms
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=3 ttl=116 time=57.1 ms
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=4 ttl=116 time=54.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 25.321/41.234/57.061/14.703 ms
pi@raspberrypi:~$
```

AT Commands:

```
+CGREG: 2,1,"B049","A014D34",7
OK
+CME ERROR: 58
+CSQ: 26,99
+QNWINFO: "TDD LTE","40440","LTE BAND 40",39150
OK
OK
OK
OK
OK
OK
OK
+QIACT: 1,1,1,"100.93.183.33"
```

AT commands:

Check SIM status: AT+CPIN?

Check Network registration: AT+CREG?, AT+CGREG?

Check signal strength: AT+CSQ

Check network type: AT+QNWINFO

Set APN: AT+CGDCONT=1,"IP","airtelgprs.com"

Activate data: AT+CGATT=1, AT+QIACT=1

Check IP address: AT+QIACT?

